

Policy considerations for effective EV-based evacuations

Improving Preparedness of Communities for Evacuations using ZEVs

3rd Stakeholder Workshop

October 2, 2025

Motivation and Aims



Inform our modeling and analysis activities



Assess and compare policy and planning strategies



Identify preferences and feasibility of different policies/planning

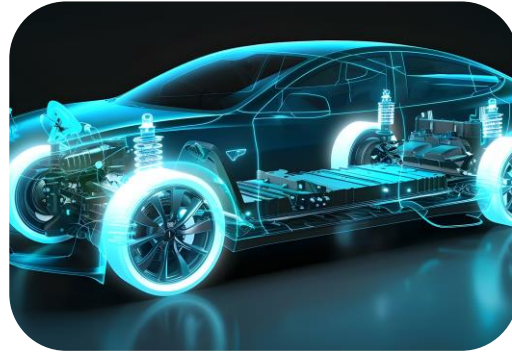


Provide guidance and recommendations for policies and practices that support positive ZEV evacuation outcomes

Policy and planning options span multiple sectors and scales



Charging
Infrastructure



Vehicles



Transportation
Infrastructure



Energy
Infrastructure

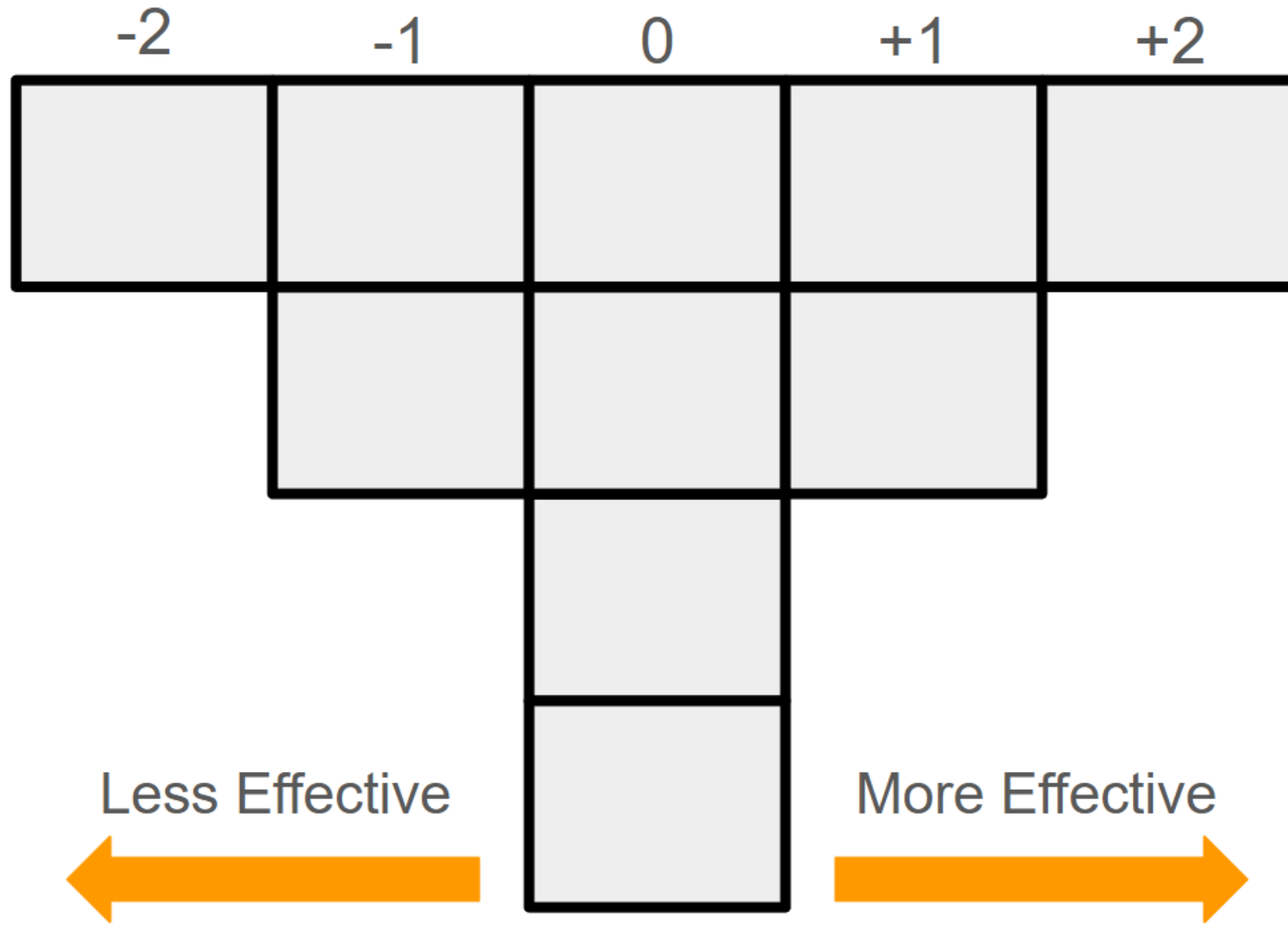
Overview of strategies for consideration

Strategy #	Strategy	Description
1	Backup energy – fossil fuels	Require public charging stations to have on-site backup energy to maintain operability during power outages. The backup power would come from diesel generators (or similar)
2	Backup energy - microgrid	Require public charging stations to have on-site backup energy to maintain operability during power outages. The backup power would come from microgrids consisting of some combination of solar/wind energy and battery energy storage
3	Free/open charging	During emergency declarations, public charging stations would be “unlocked” so that people can essentially drive up and immediately start charging without having to pay or interface with any kind of app or point of service system. Charging companies could be reimbursed by federal/state governments
4	Mobile chargers at origin	Dispatch mobile chargers (of any type) to evacuation zones prior to the event to increase the initial state of charge and reduce the need for charging along the evacuation route
5	Mobile chargers to supplement existing charging locations	Dispatch mobile charges (of any type) to sites with existing stationery charging infrastructure. The mobile chargers would act as supplemental resources to the existing charging infrastructure

Overview of strategies for consideration (continued)

Strategy #	Strategy	Description
6	Mobile chargers at designated locations along evacuation route	Dispatch mobile chargers (of any type) to designated locations (e.g., rest stops, shopping centers, etc.) along evacuation routes that have minimal existing charging infrastructure
7	Minimum “charge reserve” required on all EVs	On all EVs, require a certain % of battery charge to be kept in reserve that can be unlocked during an emergency declaration
8	Bussing/shuttling people out of evacuation zones	Use buses or shuttles (or any type) to facilitate the evacuation of people rather than relying on personal vehicles. This could reduce congestion on the road, as well as demand at downstream charging stations
9	Staggered departures from evacuation zones	Specify time periods when different locations can evacuate. This could reduce congestion on the road, as well as demand at downstream charging stations
10	Minimum “spinning reserves” required in the power grid	Require utility companies and electricity grid operators to keep a certain amount of electricity generation capacity on “standby”. This can support a rapid uptick in charging demand prior to and during an evacuation. It also provides some “backup” capacity in reduced generation from solar/wind

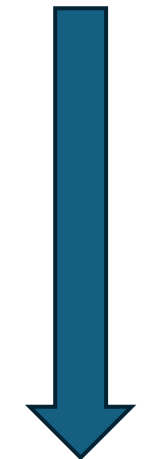
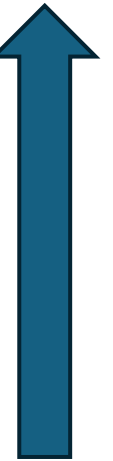
Q Sort Methodology



How do the strategies rank?

Tier 1	Mobile chargers along evacuation route
Tier 2	Free/open charging Mobile chargers to supplement existing charging
Tier 3	Mobile chargers at origin
Tier 4	Backup Energy – Fossil Fuels Minimum “charge reserve” on EVs
Tier 5	Backup Energy – Microgrid Minimum “spinning reserves” in power grid
Tier 6	Staggered departures from evacuation zones Bussing/ shuttling out of evacuation zones

More
Effective



Less
Effective

We'd love to have your input and perspective...

3rd Stakeholder Workshop - Survey on Policies and Strategies



<https://forms.office.com/r/ZxhgvxwQiC>

Policy Brief summarizing our key findings and recommendations

Preparing Communities in California for EV-Based Evacuations

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Issue

California is rapidly expanding its fleet of electric vehicles (EVs) and the charging infrastructure to support them. At the same time, recent studies highlight that climate-driven hazards in California, such as wildfires and extreme heat, are becoming more frequent and disruptive [1]. In the context of these trends, Key questions and issues arise about evacuation readiness and the appropriate number of chargers, as well as how they are distributed, managed, and prioritized during an emergency. Without integrated planning, protocols, and backup strategies, infrastructure may look sufficient on paper but fail under real-world stress. This gap between infrastructure expansion and operational readiness raises concerns about whether counties are truly prepared to support EV-based evacuations during extreme events.

Key Research Findings

*[*Note that main points are bolded with explanatory text following.]*

Educating population to keep high charge levels in EVs during hazards events is crucial.

Our research shows that electric vehicles that have high state of charge (>80%) are able to reach most shelters without needing to stop or accessing charging infrastructure [2]. California's counties are recommended to include communication campaigns to reinforce this crucial recommendation.

Strategic deployment of mobile EV chargers helps decrease evacuation time

We performed a case study in Mariposa County, CA wildfire-driven evacuation conditions with high adoption of EVs. Our results show that even modest deployment of mobile charging stations (MCS) yields significant benefits: adding 25% mobile charging capacity reduced maximum evacuation time by 33% and average wait times by 44%. Benefits arise from a compound effect, where MCS not only expands charging opportunities but also unlocks new routing patterns that improve utilization of fixed charging stations during emergencies [3].

Communities need a "playbook" for deployment of mobile EV chargers

Our consultations with communities partners reveal that some offices of emergency services do not have a clear guidelines for the logistics around mobile EV charging. Two of the critical questions include: i) who is going to finance the acquisition of these resources (local, county vs state)? ii) communities also need MOUs with possible site hosts (e.g. parking lots, public facilities, malls, etc) to deploy these chargers during emergencies.

Local officials need to be aware of the power/energy needs of EVs during emergency evacuations

During emergency evacuation, there may be a surge of power needs at charging stations. Are the communities aware of the energy and power needs during these emergencies. We recommend that local officials use tools (such as FarzanTool, see [4]) in order to gain awareness for energy/power requirements and identify gaps in the

[There is link to document in our Agenda](#)

Thank you!!!

- Please don't hesitate to reach out if you have questions, comments, suggestions, or ideas

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